

DEPI Final Project

Medical Insurance Analytics

Full Project Documentation

Team 3 — Digital Egypt Pioneers Initiative (DEPI)

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1. Project Overview

This project builds a full data-analytics pipeline on a 100,000-record medical insurance dataset — from raw CSV to cleaned, modeled data, exploratory analysis, and dashboards. The pipeline was reproduced independently across five tools: SQL, Python, Excel, Power BI, and Tableau, so results could be cross-checked against one another at every stage.

2. Team (Team 3)

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3. Dataset

Source: Medical Insurance Cost Prediction dataset, Kaggle (mohankrishnathalla/medical-insurance-cost-prediction).

Size: 100,000 rows, 54+ raw columns (56 after cleaning and feature engineering).

Unit of observation: one insured individual per row.

Variable Categories

Category	Example Columns
Demographics & Socioeconomic	age, sex, region, income, education, employment_status
Lifestyle & Habits	bmi, smoker, alcohol_freq
Health & Clinical Indicators	hypertension, diabetes, cardiovascular_disease, hba1c, risk_score
Healthcare Utilization & Procedures	visits_last_year, hospitalizations_last_3yrs, proc_surgery_count
Insurance & Policy Information	plan_type, network_tier, deductible, provider_quality
Medical Costs & Claims	annual_medical_cost, annual_premium, claims_count, total_claims_paid

Full column-by-column definitions are provided in DOCUMENTS/Medical Insurance Dataset Description.pdf in the repository.

4. Repository Structure

Folder	Contents
RAW DATA/	Original, unmodified medical_insurance.csv from Kaggle
SQL/	MySQL scripts: database setup & cleaning, star-schema modeling, business analysis
Python/	Jupyter notebook covering cleaning, modeling, EDA, and dashboards
EXCEL/	Power Query cleaning/modeling workbook, cleaned CSV export, dashboard workbook
Power BI/	.pbix report with cleaning, modeling, and a multi-page report
Tableau/	.twbx workbook with the final visualization dashboards
DOCUMENTS/	Dataset description and team member/role documentation

5. Workflow

The project follows the same five-stage pipeline in every tool that supports it:

1. Raw CSV (Kaggle)
2. Data Cleaning — done independently in SQL, Python, Excel, and Power BI
3. Star-Schema Modeling — same four tools, same schema
4. Analysis — SQL queries and Python EDA
5. Dashboards — Excel, Power BI, Python, and Tableau

6. Data Cleaning

Cleaning was performed independently in every tool that supports it — SQL, Python, Excel (Power Query), and Power BI (Power Query) — so results could be cross-checked across tools. Tableau was used purely for visualization on top of the already-cleaned dataset, since it isn't built for row-level data cleaning.

Issues Identified & Corrections Applied

1. Implausible ages

Rows with age under 16 or age 95 and above were flagged as suspicious when other attributes contradicted the reported age — for example, still employed, very few medications on file, no chronic conditions, or an active smoker at an extreme age. Flagged rows were corrected, not dropped, using a cascading contextual median:

- 1st choice: median age within the same group of education, employment status, marital status, alcohol frequency, and smoking status

- 2nd choice (fallback): median within a 4-factor group (drops smoking status)
- 3rd choice (fallback): global median across all valid rows

The result is always floored at a rule-based minimum plausible age for that record (e.g. a current smoker can't be younger than 16; a retiree can't be younger than 55).

2. Diabetes / HbA1c inconsistency

Records marked diabetes = 1 but with an HbA1c lab value below the 6.5% clinical diagnostic threshold were corrected to diabetes = 0, since the lab result doesn't support the diagnosis.

3. Procedure mismatch

Records marked had_major_procedure = 1 with proc_surgery_count = 0 were corrected to had_major_procedure = 0.

4. Missing lifestyle data

Missing alcohol_freq values were filled as "Unknown" rather than dropped.

5. Feature engineering (Excel pass)

- income_log — log-transformed income, for less skewed distribution analysis
- flag_high_cost_no_claims — flags members with high medical cost but no recorded claims

7. Data Modeling — Star Schema

The same star schema was rebuilt in SQL, Python, Excel (Power Query), and Power BI:

Table	Role
fact_insurance	Utilization, risk score, costs, premiums, claims, and procedure counts
dim_person_info	Core person attributes
dim_socioeconomic	Income, education, employment, household
dim_lifestyle	BMI, smoking, alcohol frequency
dim_chronic	Chronic condition flags and count
dim_clinical	Blood pressure, LDL, HbA1c, risk score
dim_policy	Deductible, copay, policy term, provider quality
dim_plan_type / dim_network_tier	Lookup tables referenced by dim_policy

8. Analysis

SQL and Python analysis covered ten categories of business questions:

- KPIs and customer segmentation (age, gender, urban vs. rural, region)

- Lifestyle analysis (smoking, BMI category, alcohol consumption)
- Clinical risk analysis (blood pressure, diabetes, LDL)
- Chronic disease analysis (cost by condition count, top cost-driving conditions, multi-condition members)
- Policy analysis (plan profitability, network tier claims, deductible impact)
- High-risk member analysis and risk segment distribution
- Utilization analysis (most-used procedures, hospitalization impact)
- Socioeconomic analysis (income vs. risk score, education level)
- Business insights (top customers by cost, loss ratio by plan/risk segment, smoking impact on claims)

The Python notebook additionally covers 30+ exploratory breakdowns, including cost deciles, cumulative cost share, and correlation analysis of every factor against `annual_medical_cost`.

9. Dashboards

Tool	Pages / Dashboards
Excel	Demographics, Customers, Risk Analysis, plus a Data Model view
Power BI	Home, Executive Overview, Financial Performance, Claims & Healthcare Utilization, Population Health & Risk, Members Demographics & Policy
Python	Insurance Cost Overview, Risk Factors Analysis, Financial & Risk Segmentation Analysis, Executive Insights
Tableau	Demographics, Customers, Risk Analysis, Policy & Operations

Dashboard screenshots are hosted in the GitHub repository (not embedded here) — see the tool-specific sections below for direct links.

10. How to Reproduce

1. Start from RAW DATA/medical_insurance.csv.
2. Run the cleaning step in your tool of choice (SQL/final project sql cleaning commented.sql, or the "cleaning" section of Python/DEPI_insurance_project.ipynb, or the Excel/Power BI Power Query steps).
3. Build the star schema (SQL/final project sql data model commented.sql, or the modeling section of the notebook).
4. Run the SQL analysis script or the Python EDA cells for the business questions above.
5. Open the dashboard files directly: EXCEL/DATA MODEL + DASHBOARD.xlsx, Power BI/Power BI Final Project.pbix, or Tableau/tableau final project fi.twbx.

11. SQL Phase — Database Setup, Cleaning, Modeling & Analysis

Built and run in MySQL Workbench. This phase covers the full pipeline from raw table to a queryable star schema, in three scripts.

Files

File	Purpose
final project sql cleaning commented.sql	Creates the database, validates and corrects implausible ages
final project sql data model commented.sql	Builds the star schema (fact + dimension tables) from the cleaned data
final project SQL Analysis.sql	Runs the business analysis queries against the star schema

Cleaning

Implausible ages are flagged and corrected using the same cascading median-imputation strategy described in Section 6. Because MySQL doesn't support UPDATE ... FROM, the medians are computed via window functions and joined back using MySQL's multi-table UPDATE ... JOIN syntax.

Data Model

Same star schema as Section 7: fact_insurance plus dim_person_info, dim_socioeconomic, dim_lifestyle, dim_chronic, dim_clinical, and dim_policy (referencing dim_plan_type and dim_network_tier).

Analysis

Answers all ten categories of business questions listed in Section 8.

Folder link: <https://github.com/nourhanelboraey/DEPI-Final-project-Medical-Insurance/tree/main/SQL>

12. Python Phase — Cleaning, Modeling, EDA & Dashboards

Notebook: DEPI_insurance_project.ipynb (pandas, matplotlib/seaborn). This phase independently reproduces the full pipeline in Python as a cross-check against the SQL and Excel versions.

Cleaning & Modeling

Same corrections and star schema as Sections 6 and 7, implemented with pandas.

Exploratory Data Analysis

- Summary statistics (cost, BMI, age range) and cost by smoking status, region, sex, and chronic condition count
- Age-group and cost-bucket segmentation
- Regional cost distribution vs. regional average, and top-5 costs by region

- Correlation of age, risk profile, and every numeric factor against annual_medical_cost
- Cost decile analysis and cumulative share of total cost
- High-risk vs. low-risk comparison (cost, claims, chronic conditions)
- Premium vs. cost by risk level, and loss ratio by plan type
- Procedure count correlation with cost, and profile of the top 10% most expensive members

Dashboards

1. Insurance Cost Overview — total costs, members, premiums, average cost per member, cost distribution
2. Risk Factors Analysis — smoking, BMI, and chronic condition impact on cost
3. Financial & Risk Segmentation Analysis — loss ratio, average premium, plan types, high-risk member counts
4. Executive Insights — highest-cost members, premium vs. cost comparison, and top-line risk distribution

Folder link: <https://github.com/nourhanelboraey/DEPI-Final-project-Medical-Insurance/tree/main/Python>

13. Excel Phase — Cleaning, Modeling & Dashboards

Files

File	Purpose
POWER QUERY.xlsx	Power Query cleaning steps and star-schema table build
medical_insurance_cleaned.csv	Cleaned dataset exported for use in the other tools
DATA MODEL + DASHBOARD.xlsx	Data model relationships and the final dashboards

Cleaning (Power Query)

Mirrors the logic used in SQL and Python (Section 6). Two engineered columns were added at this stage: income_log (log-transformed income) and flag_high_cost_no_claims (flags high-cost members with no recorded claims).

Dashboards

DATA MODEL + DASHBOARD.xlsx contains the Demographics, Customers, and Risk Analysis dashboards, plus a Data Model view showing the relationships between the fact and dimension tables.

Folder link: <https://github.com/nourhanelboraey/DEPI-Final-project-Medical-Insurance/tree/main/EXCEL>

14. Power BI Phase — Cleaning, Modeling & Report

File: Power BI Final Project.pbix

Cleaning & Transformation

Rebuilt independently in Power Query (M), applying the same corrections used in the other tools (Section 6).

Data Model

Same star schema as Section 7, with DAX measures built on top for metrics such as smoker-status cost comparisons and chronic-condition impact.

Report Pages

1. Home
2. Executive Overview
3. Financial Performance
4. Claims & Healthcare Utilization
5. Population Health & Risk
6. Members Demographics & Policy

Folder link: <https://github.com/nourhanelboraey/DEPI-Final-project-Medical-Insurance/tree/main/Power%20BI>

15. Tableau Phase — Visualization & Dashboards

File: tableau final project fi.twbx

Tableau was used purely for visualization, built on top of the dataset already cleaned and modeled in SQL, Python, Excel, and Power BI — Tableau isn't intended for row-level data cleaning, so no cleaning logic lives in this phase.

Dashboards

1. Demographics
2. Customers
3. Risk Analysis
4. Policy & Operations

Folder link: <https://github.com/nourhanelboraey/DEPI-Final-project-Medical-Insurance/tree/main/Tableau>

Full source files, notebooks, and dashboard screenshots are available in the GitHub repository linked on the title page.